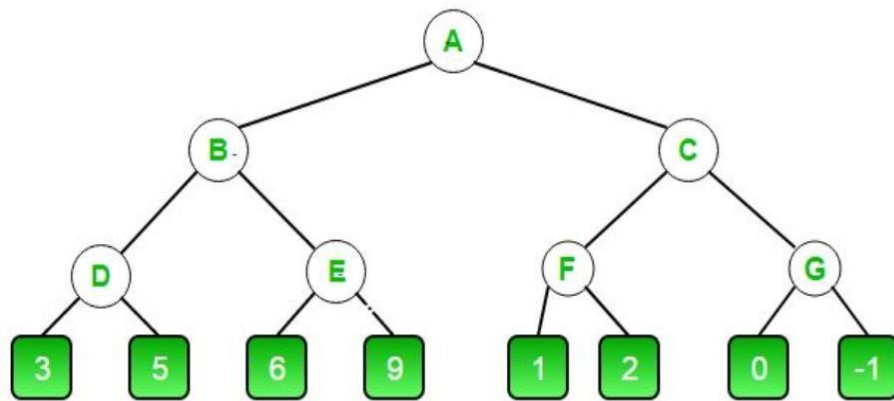


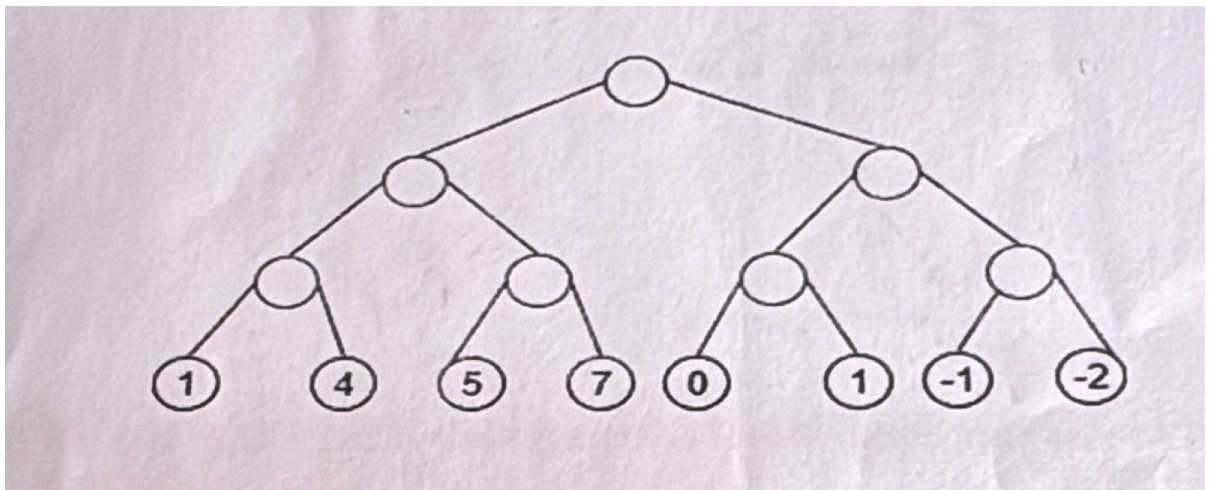
CHAPTER 4

PRACTICE PROBLEMS

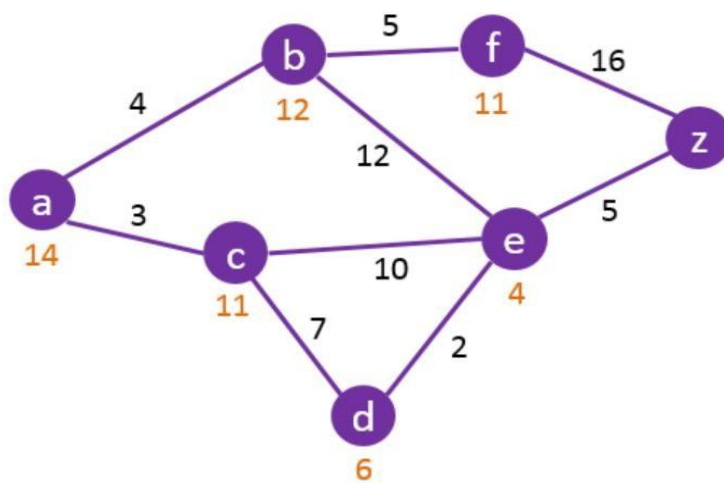
1. In state space search, heuristics are formalized as rules for choosing those branches in a state space that are most likely to lead to an acceptable problem solution. AI problem solvers employ heuristics in two basic situations, what are those situations?
OR State the problem solving scenarios wherein heuristic are applicable?
2. Why are simple games such as the 8-puzzle ideal for exploring the design and behavior of heuristic search algorithms?
3. When we say a search is admissible what do we mean?
4. When do we say a search is informed and what do we mean?
5. State the property of Monotonicity
6. Explain the equation of A* Search Algorithm
7. Use alpha-beta pruning algorithm to find a solution that can lead to winning a game and the path that leads to winning a game.



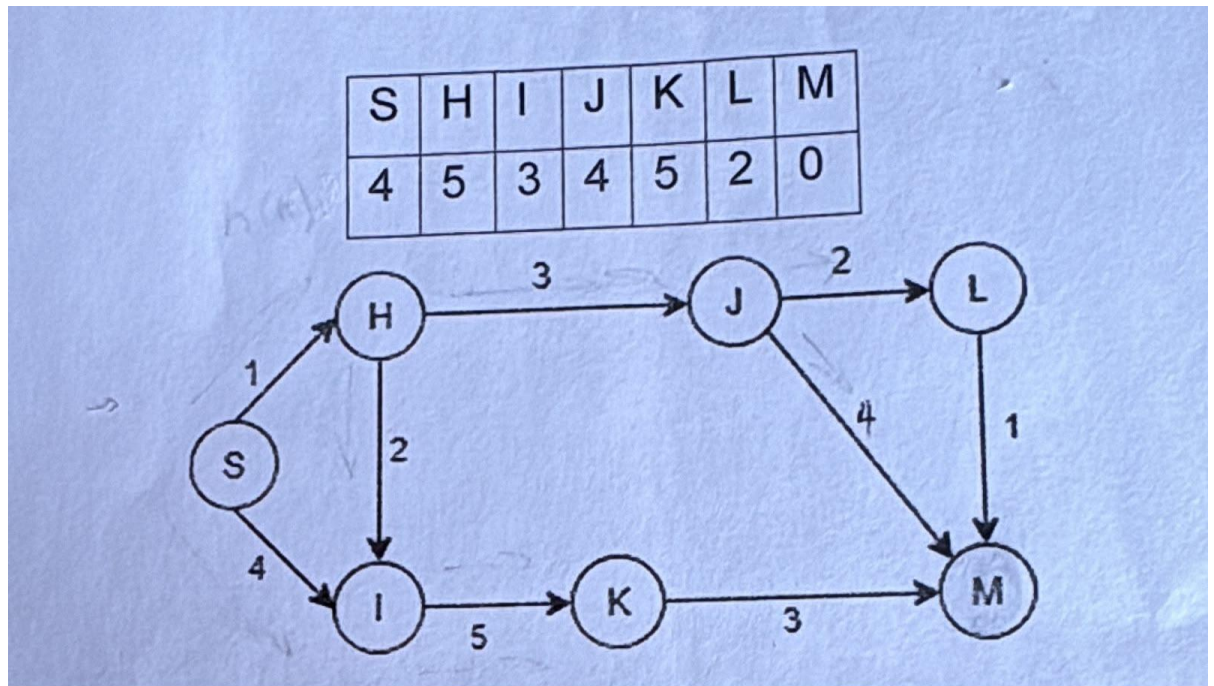
8.



9. Use A* algorithm to find a path from a to z.

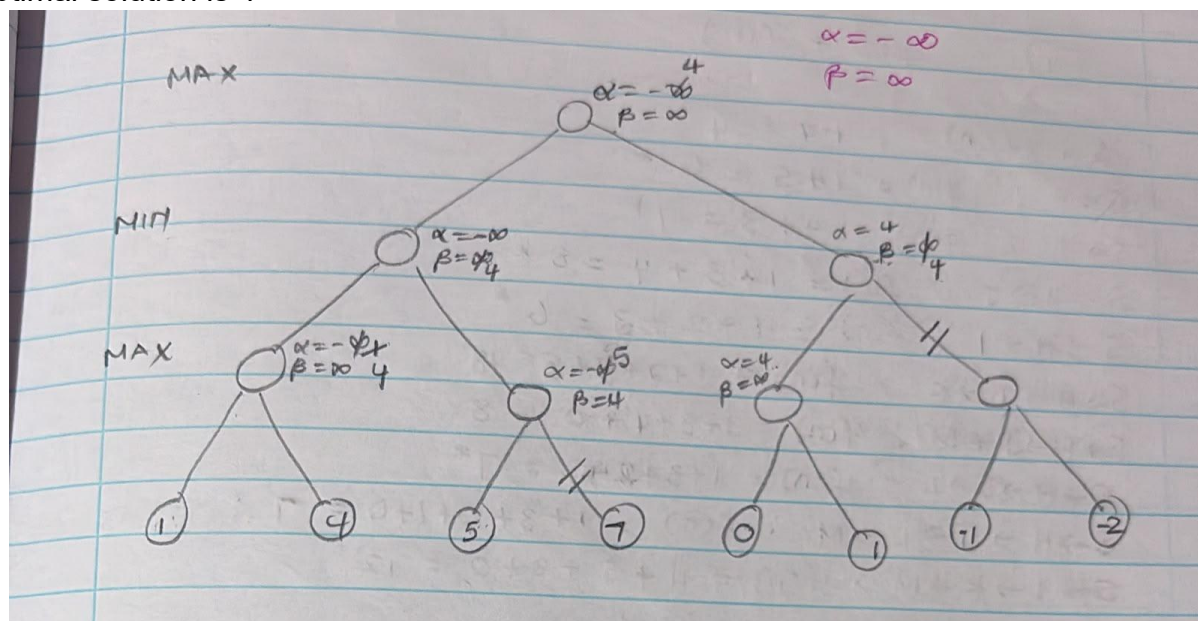


10.



Solutions

1. Page 123
2. Page 143
3. A search algorithm is admissible if it is guaranteed to find a minimal path to a solution whenever such a path exists.
7. Optimal solution is 5
8. Optimal solution is 4



9. $F(n) = g(n) + f(n)$

a: $f(n) = 0 + 14 = 14$

a -> b: $f(n) = 4 + 12 = 16$

a -> c: $f(n) = 3 + 11 = 14^*$

a -> c -> e: $f(n) = (3 + 10) + 4 = 17$

a -> c -> d: $f(n) = (3 + 7) + 6 = 16^*$

a -> c -> d -> e: $f(n) = (3 + 7 + 2) + 4 = 16^*$

a->b->f: $f(n) = 4 + 5 + 11 = 20$

a->b->e: $f(n) = 4 + 12 + 4 = 20$

a -> c -> d -> e -> b: $f(n) = (3 + 7 + 2 + 12) + 12 = 36$

a -> c -> d -> e -> z: $f(n) = (3 + 7 + 2 + 5) + 0 = 17$

Therefore, total cost is 17 and the path is a -> c -> d -> e -> z

10. $f(n) = g(n) + h(n)$

s: $f(n) = 0 + 4 = 4$

s->h: $f(n) = 1 + 5 = 6^*$

s->i: $f(n) = 4 + 3 = 7^*$

s->h->j: $f(n) = 1 + 3 + 4 = 8^*$

s->h->i: $f(n) = 1 + 2 + 3 = 6^*$

s->h->i->k: $f(n) = 1 + 2 + 5 + 5 = 13$

s->h->j->m: $f(n) = 1 + 3 + 4 + 0 = 8$

s->h->j->l: $f(n) = 1 + 3 + 2 + 2 = 7^*$

s->h->j->l->m: $f(n) = 1 + 3 + 2 + 1 + 0 = 7$

s->i->k->m: $f(n) = 4 + 5 + 3 + 0 = 12$

Path: s->h->j->l->m

Path cost = 7